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Hybridisation and pi-system reasoning — Issue #35

independent candidate: Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: From Lewis contributors to one delocalised pi description

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Formal contributors to delocalised orbital model

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Nitrate, NO_3^- , is planar and has three equivalent N–O bonds. Its three conventional Lewis contributors place one

Representation

N=O bond in different positions. Choose between a permanently localised single/double-bond picture and a delocalised pi description. Explain which observations discriminate between them, and state separately the introductory sigma-framework hybridisation model at nitrogen.

Common wrong route • open if you want a warning

+

Supplied nitrate contributors



Contributor 1

Contributor 2

Cont

Connectivity layout only; page angles are not measured geometry.

Supplied situation

Need a hint? • guided support

+

Need the concept again?

From Lewis contributors to one delocalised pi description

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: From Lewis contributors to one delocalised pi description

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Formal contributors to delocalised orbital model

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Use the supplied nitrate description: three planar sigma bond directions around nitrogen and a pi system delocalised across the N–O framework. Translate the three Lewis contributors into one qualitative orbital diagram. Distinguish the sigma framework from the delocalised pi system, and explain what is preserved and what must not be interpreted as a molecule switching between drawings.

Representation

Nitrate contributors to be translated



Contributor 1

Contributor 2

Cont

Connectivity layout only; page angles are not measured geometry.

Common wrong route · open if you want a warning

+

Supplied situation

Need a hint? · guided support

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Need the concept again?

From Lewis contributors to one delocalised pi description

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Coordinate p-orbital directions across several centres

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: P-orbital orientation and continuity

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For propadiene (allene), $\text{H}_2\text{C}=\text{C}=\text{CH}_2$, apply local electron-domain counting at each of the three carbon atoms. Assign the

Representation

introductory hybridisation of each carbon and show how the two mutually perpendicular unhybridised p orbitals on the central carbon constrain the relative planes of the terminal CH₂ groups.

Common wrong route · open if you want a warning

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Supplied allene connectivity



Allene connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Coordinate p-orbital directions across several centres

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Coordinate p-orbital directions across several centres

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: P-orbital orientation and continuity

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Ethene is planar; propadiene has terminal CH₂ groups in perpendicular planes. Build a connected account from each carbon's local sigma framework, the required pi overlaps and the p-orbital directions. Explain why copying the planar ethene sketch to allene fails.

Representation

Common wrong route · open if you want a warning

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Ethene and allene connectivity



Ethene connectivity



Allene connectivity

Connectivity layout only; page angles are not measured geometrically
Supplied situation

Need a hint? · guided support

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Need the concept again?

Coordinate p-orbital directions across several centres

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: A reliable atom-local hybridisation procedure

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Atom-local hybridisation procedure

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For the benchmark, use these supplied simplified observations, not an exam-source claim: CO₂ is linear with two equivalent C–O bonds; NO₃[–] is planar with three equivalent N–O bonds; NH₃ is pyramidal with three equivalent N–H bonds. Assess the claim 'Equivalent bonds imply sp² at the central atom.' Use counterexamples and valid warrants, and distinguish equivalence, electron-domain

Representation

count and hybridisation as a model description.

Common wrong route · open if you want a warning

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Supplied equivalence observations

CO₂: equivalent C-O; supplied linear shape

NO₃⁻: equivalent N-O; supplied planar shape

NH₃: equivalent N-H; supplied pyramidal shape

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

A reliable atom-local hybridisation procedure

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: A reliable atom-local hybridisation procedure

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: Atom-local hybridisation procedure

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For propyne, $\text{CH}_3\text{--C}\equiv\text{CH}$, assign the introductory hybridisation of each carbon and count the sigma and pi bonds in the entire molecule. Show a check that distinguishes the triple bond's components from the local domain count.

Representation

**Common wrong route · open if you
want a warning**

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Supplied propyne connectivity



Propyne connectivity

Connectivity layout only; page angles are not meas

Supplied situation

Need a hint? · guided support

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Need the concept again?

A reliable atom-local hybridisation procedure

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: From Lewis contributors to one delocalised pi description

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Formal contributors to delocalised orbital model

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies that carbonate, CO_3^{2-} , is planar and has three equivalent C–O bonds; its three conventional Lewis contributors place one C=O bond in different positions. Construct one qualitative description of its sigma framework and delocalised pi system. State

Representation

why no single contributor should be read as the complete physical picture.

Common wrong route · open if you want a warning

+

Supplied carbonate contributors



Contributor 1

Contributor 2

Cont

Connectivity layout only; page angles are not measured geometry.

Supplied situation

Need a hint? · guided support

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Need the concept again?

From Lewis contributors to one delocalised pi description

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Coordinate p-orbital directions across several centres

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: P-orbital orientation and continuity

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Compare rotation about the central C–C bond in ethane with twisting about the C=C bond in ethene. Use the sigma and pi overlap arrangements to explain which overlap is altered by each motion. Do not infer a numerical energy barrier from a sketch.

Common wrong route · open if you want a warning

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Representation

Ethane and ethene connectivity



Ethane

H₂C

Ether

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Coordinate p-orbital directions across several centres

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Coordinate p-orbital directions across several centres

Concept difficulty: Hard concept

Question difficulty: Connected reasoning

Family: P-orbital orientation and continuity

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ in an approximately planar conformation, identify the continuous sequence of p orbitals and contrast it with $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$. Explain which structural feature interrupts that pathway in the second molecule.

Representation

**Common wrong route · open if you
want a warning**

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Supplied pathway comparison



First connectivity



Second connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Coordinate p-orbital directions across several centres

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: A reliable atom-local hybridisation procedure

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Atom-local hybridisation procedure

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student assigns one hybridisation label to an entire molecule. Use propyne and allene to test this procedure and devise a more reliable local procedure. State the electron-domain information and orbital-consistency checks your procedure needs.

Representation

Common wrong route · open if you want a warning

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Supplied local-procedure tests



Propyne connectivity



Allene connectivity

Connectivity layout only; page angles are not measured geometry

Supplied situation

Need a hint? · guided support

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Need the concept again?

A reliable atom-local hybridisation procedure

Chemistry · Hybridisation and pi-system reasoning — Issue #35 independent candidate